

ROBERVAL, QC, Canada

WMO: 717280

Lat: 48.5230N

Lon: 72.2662W

Elev: 586

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-17.0	-32.2	0.9	-21.1	-27.7	1.2	-16.3	29.3	9.6	25.9	10.0	7.3	250	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	83.1	68.1	79.9	66.6	76.6	65.2	70.5	79.3	68.7	76.8	67.0	74.5	10.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	102.8	75.5	65.6	96.4	73.6	63.8	90.6	71.7	34.8	79.4	33.3	76.8	31.9	74.5	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	20.8	18.5	DB	-28.3	89.7	5.5	3.7	-32.3	92.3	-35.6	94.5	-38.7	96.6	-42.7	99.2	(4)
(5)			WB	-28.7	73.9	5.4	1.9	-32.6	75.3	-35.7	76.5	-38.8	77.6	-42.7	79.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	4.5	8.4	20.3	35.0	50.3	60.7	65.5	63.4	55.4	42.8	29.5	14.3		
	DBStd	23.28	14.01	13.04	12.37	9.00	8.57	7.70	5.75	5.95	7.93	8.16	9.81	12.42		
	HDD50	6065	1409	1164	921	453	102	7	0	0	32	253	618	1106		
	HDD65	10172	1874	1584	1385	899	463	172	64	103	304	687	1066	1571		
	CDD50	1565	0	0	1	4	112	329	479	414	194	31	1	0		
	CDD65	197	0	0	0	0	7	44	78	52	15	1	0	0		
	CDH74	1293	0	0	1	2	95	347	470	286	88	4	0	0		
	CDH80	273	0	0	0	0	22	87	97	52	15	0	0	0		
	WSAvg	8.4	8.6	8.9	9.3	8.9	8.1	7.8	7.5	7.7	8.1	8.5	8.9	8.5		
	PrecAvg	34.0	2.2	1.9	1.9	2.4	2.5	3.4	4.2	3.5	3.7	3.2	2.6	2.6		
(15) Precipitation	PrecMax	43.3	5.8	4.6	3.8	4.6	5.0	5.5	9.8	9.8	8.5	6.2	4.3	4.8		
	PrecMin	27.1	0.5	0.7	0.7	0.6	0.5	1.5	1.7	1.4	1.5	0.6	0.5	0.7		
	PrecStd	4.1	1.1	0.9	0.9	0.9	1.0	1.1	1.9	1.7	1.5	1.3	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	43.6	52.9	65.2	83.0	87.3	87.0	85.1	81.7	71.4	57.9	42.6		
		MCWB	42.7	39.0	43.1	50.5	62.6	69.1	69.9	70.0	69.0	59.4	52.4	40.7		
	2%	DB	36.3	36.0	44.6	58.5	75.7	82.2	82.6	80.8	76.1	64.9	51.3	37.7		
		MCWB	34.7	33.1	37.8	46.7	59.2	66.1	68.7	68.0	66.5	56.9	47.1	35.4		
	5%	DB	31.2	32.4	40.3	52.6	70.2	77.8	79.2	77.0	71.7	59.8	46.5	34.1		
		MCWB	29.5	29.9	35.3	43.2	56.7	64.7	67.0	66.2	64.1	53.4	42.5	32.2		
	10%	DB	25.3	28.5	36.8	47.6	65.6	73.7	75.9	73.7	68.2	55.2	42.0	31.7		
		MCWB	23.6	26.2	32.9	40.1	54.2	62.2	65.7	64.6	61.3	50.2	38.9	30.2		
	0.4%	WB	42.7	40.2	46.4	54.3	65.9	72.1	73.4	72.0	70.6	61.7	53.1	40.3		
		MCDB	44.6	43.0	48.5	62.3	78.0	83.4	82.0	81.9	78.6	68.4	57.0	41.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.0	33.4	38.7	47.5	62.1	68.9	70.7	69.5	67.8	57.5	47.8	35.6		
		MCDB	36.3	35.8	43.4	55.5	71.2	78.1	78.9	78.3	74.5	64.3	50.9	37.4		
	5%	WB	29.4	30.1	35.8	44.1	58.2	66.6	68.8	67.7	64.9	54.0	43.0	32.6		
		MCDB	31.2	32.5	39.6	51.5	67.8	75.2	77.1	74.7	71.2	58.6	45.9	33.9		
	10%	WB	23.6	26.2	33.1	40.9	54.9	64.1	66.8	65.8	62.1	50.6	39.2	30.3		
		MCDB	25.3	28.6	36.6	47.0	64.4	72.0	74.5	72.2	67.3	54.4	41.7	31.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	17.2	17.2	16.4	19.8	18.5	16.7	16.6	16.5	13.7	12.1	13.9		
		MCDBR	20.5	17.0	18.7	24.2	27.2	22.6	20.1	20.2	20.4	20.2	17.5	14.3		
		MCWBR	19.8	15.4	14.5	15.7	14.9	12.7	11.0	11.5	13.0	13.9	14.7	13.8		
		MCDBR	20.4	16.8	17.0	21.8	23.8	19.8	17.9	17.4	18.7	18.3	16.9	14.3		
	5% WB	MCWBR	19.9	15.5	13.9	15.3	14.7	12.4	10.6	10.9	13.0	13.6	15.0	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.280	0.303	0.371	0.352	0.383	0.395	0.374	0.362	0.329	0.314	0.259		
	taud		2.235	2.225	2.238	2.257	2.431	2.367	2.352	2.413	2.432	2.505	2.372	2.318		
	Ebn at Noon		252	277	289	278	288	279	273	275	267	259	233	243		
	Edh at Noon		24	31	36	40	35	37	37	34	30	24	22	19		
(44) All-Sky Solar Radiation	RadAvg		420	732	1148	1449	1650	1773	1637	1403	1073	617	334	274		
	RadStd		54	95	101	122	153	128	163	89	87	61	39	32		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	+3.02	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page